

Melissa Agbaje

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CAREER PROFILE

Results-driven Analytics and Mathematics professional with dual bachelor's degrees in Accounting and Mathematics and an ongoing Master's in Advanced Data Analytics. Combines strong quantitative and financial foundations with hands-on experience in data analysis, predictive modeling, and technical problem-solving. Proficient in Python, R, SQL, and C++, with exposure to cloud platforms, data visualization tools, and enterprise software. Proven ability to communicate complex findings clearly, work across technical and non-technical environments, and deliver actionable insights that drive informed decision-making.

EDUCATION

Eligible CPA Candidate | 150 Credit Hours

University of North Texas
Master of Science in Advanced Data Analytics | GPA: 4.0
Relevant Coursework: Data Analytics, Discovery and Learning with Big Data, Biostatistics, Biocomputing, Time Series Econometrics and Forecasting, Contemporary Mathematics

Denton, TX
Expected: May 2027

University of North Texas
Bachelor of Business Administration in Accounting | GPA: 3.3
Relevant Coursework: Accounting Information Systems, Business Communication, Advanced Business Financial Analysis, Strategic Management, Federal Income Taxation, Data Mining

Denton, TX
May 2023

University of North Texas
Bachelor of Science in Mathematics | GPA: 3.4
Relevant Coursework: Statistics, Probability, Mathematical Modeling, Data Analysis, Vector Calculus, Computer Science and Engineering, Multivariable Calculus

Denton, TX
May 2023

EXPERIENCE

CliftonLarsonAllen LLC
Tax Intern

Dallas, TX
January 2024 – April 2024

- Prepared 10+ complex tax returns weekly (1120S, 1120, 1065, 1040, 1041) using CCH Axcess, maintaining accuracy across high-volume client workloads.
- Organized and analyzed client financial documents to build supporting workpapers, streamlining the tax preparation workflow.
- Developed working knowledge of federal tax compliance procedures and systems through hands-on client work, peer collaboration, and self-directed study.

University of North Texas
Physics Student Assistant and Lab Assistant

Denton, TX
September 2021 – May 2023

- Supported 2 faculty members in course preparation and academic operations across a multi-semester engagement spanning nearly two years.
- Evaluated 100+ student assignments and quizzes weekly, providing consistent and timely feedback to support academic performance.
- Redesigned the university's physics research and demonstrations website, improving accessibility and usability for faculty and students.
- Maintained and organized laboratory equipment inventory to ensure reliable setup and execution of class demonstrations.

PROJECTS

Vehicle Safety Complaint Classification | Python, TensorFlow, Keras, Scikit-learn, Pandas
University of North Texas — Advanced Data Analytics Coursework

- Cleaned and preprocessed a dataset of 418,000+ real-world NHTSA vehicle safety complaints, handling missing values, duplicate records, and multi-format date fields across 49 raw variables.
- Built and trained an LSTM deep learning model to classify high-risk complaints (crashes, fires, injuries, deaths), achieving 98% accuracy and an AUC of 0.968 on a held-out test set of 83,000+ records.
- Developed a CNN text classification model that outperformed the LSTM on recall for high-risk cases (0.87 vs. 0.83), demonstrating stronger sensitivity in a safety-critical imbalanced dataset (7% positive class).
- Extended the pipeline to a 10-class multi-label component prediction task (brakes, engine, airbags, etc.) across 304,000 complaints, achieving 59% accuracy using an LSTM with sparse categorical cross-entropy loss.

- Visualized model performance using confusion matrices, AUC bar charts, and training/validation accuracy curves to support model selection and interpretation.

Nonprofit Donor Classification & Gift Amount Prediction | Python, Scikit-learn, Pandas, Matplotlib

University of North Texas — Advanced Data Analytics Coursework

- Designed a dual-model predictive pipeline on 6,002 donor records to solve a real-world direct mail profitability problem where mass mailing produced a loss of \$0.55 per recipient.
- Built and compared 3 classification models (Naive Bayes, KNN, Random Forest) to predict donor likelihood; selected KNN (k=10) for highest recall (0.93) and expected profit of \$6.31, prioritizing revenue recovery over false-positive cost.
- Built and compared 3 regression models (Linear Regression, KNN, Gradient Boosting) to predict donation amounts; selected Gradient Boosting for lowest RMSE (\$1.14) and highest R² (0.66).
- Deployed the combined pipeline on 2,007 unseen records, reducing the mailing list from 2,007 to 636 targeted recipients and flipping campaign outcome from a -\$1,104 loss to \$7,768 in expected profit — an \$8,872 improvement.

Missing Data Treatments — Technical Analysis | Python, Statistics

University of North Texas — Advanced Data Analytics Coursework

- Conducted an in-depth technical study of missing data mechanisms (MCAR, MAR, MNAR) and their statistical implications on analytical model integrity.
- Evaluated and compared multiple missing data treatment methods including listwise deletion, mean/mode imputation, and multiple imputation techniques, analyzing trade-offs in bias, variance, and data loss.
- Presented findings to faculty and peers, communicating complex statistical concepts and method selection rationale in a clear, structured format.
- Demonstrated how improper handling of missing data can distort model outputs, reinforcing the importance of treatment selection in real-world data pipelines.

SKILLS

Technical: Google Cloud Platform, Amazon Web Services, CCH Axxess Tax, CCH ProSystem Fx Engagement, Microsoft Office Suite (Excel, Outlook, PowerPoint, Word), Tableau, Linux, Hive, Spark, Power BI

Programming: Python, C++, R, MATLAB, SQL, Alteryx

Professional: Communication, Teamwork, Attention to Detail, Problem Solving, Analytical Solver

Languages: English (Fluent), Yoruba, Igbo

ACTIVITIES AND LEADERSHIP

The Honor Society of Phi Kappa Phi | Member

March 2026

AWARDS AND HONORS

Dean's List | **GPA 3.8**

Fall 2020, Spring 2021

University of North Texas — Denton, TX